

```
FUNCTION TrapezoidalRule(f, a, b, n)
```

```
  h <- (b - a) / n
```

```
  sum <- 0.5 * (f(a) + f(b))
```

```
  FOR i FROM 1 TO n-1
```

```
    x <- a + i*h
```

```
    sum <- sum + f(x)
```

```
  END FOR
```

```
  RETURN h * sum
```

```
END FUNCTION
```

```
FUNCTION main()
```

```
  INPUT f, a, b, n
```

```
  OUTPUT approxIntegral <- TrapezoidalRule(f, a, b, n)
```

```
END main()
```